



The Store Age

Upcoming storage technologies to keep an eye on

SACD

The Super Audio CD (SACD) manufactured by both, Sony and Philips Electronics, can accommodate more than four times the information of the current CD format. The extra capacity provides space for two-channel stereo data, an area for up



to six-track multi-channel data, storage capacity for text and images, disc variations, copyright protection, et al.

Launched in 1999, SACD is a new approach to audio recording and reproduction. The extra storage capacity and better sound quality are likely to make SACDs a rage among music aficionados, in the coming months.

SACD uses Direct Stream Digital (DSD) recording technology to record and playback results, at a frequency response of over 100 KHz, and a dynamic range of over 120 dB across the audible frequency range. DSD increases the resolution of music by following its original waveform more closely. This results in music reproduction that's astonishingly true to the original.

Super Audio CD players can playback SACDs and all the current CDs. However, normal CD players can only play hybrid SACDs—those containing a CD layer.

SACD captures sound at the full DSD bit rate, with DSD quality. Producers may also use the additional channels to faithfully reproduce the precise acoustic signature of the performance space.

Dual layered DVDs

HP, Dell and Philips will start marketing their double-layer DVD recording devices in a few months. Discs on these systems are expected to hold 8.5 GB—that's four hours of DVD-quality video, or 16 hours of VHS-quality video. These companies are

also working on dual-layer recording, which involves writing data on two separate surfaces of a DVD.

Single-layer recordable DVDs can hold up to 4.7 GB of data. Retail DVD movies have the data written on two layers, but the discs are stamped, not burnt with a laser, as DVD-Writers do.

Dual-layer recording has not been available to the average consumer in the past. Philips' Optical Storage division has announced that their dual-layer DVD recorders slated for PC use will hit the markets later this year, along with dual-layer recorders targeted at the consumer electronics market.

Paper Discs

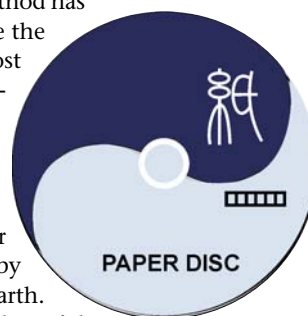
Storage has gone from floppies to zip discs, CDs, DVDs and now, paper discs! Sony and Toppan Printing have announced a 25 GB Blu-ray disc—51 per cent of the disc's weight is paper. Data can be printed directly on the disc, and it can store more than two hours of high-definition video.

The paper discs are easily disposable—you can shred them with a scissors. They are also environment-friendly, as they use an easily recyclable resource.

The Blu-ray disc format uses a blue laser, and is considered the current red laser technology's successor. Blu-ray discs store more than five times the capacity of current DVDs—that's 25 GB of data!

Discs using organic materials are also being developed by other companies: Sanyo's MildDisc, launched in early April, is made from polyactic acid derived from corn kernels. The disc is currently only available as a CD. This method has been developed to replace the polycarbonate used in most discs today. Also, polycarbonate discs aren't biodegradable.

According to Sanyo, their MildDisc can be broken down into water and carbon dioxide by micro-organisms in the earth. They also claim that no harmful dioxins are released during incineration, which can be achieved at much lower temperatures than for polycarbonate discs. 



PEOPLE TO WATCH

At 29, both Larry Page and Sergey Brin, directors of Google Inc, have the Web revolving round them. After having changed the way the world uses the Internet, they have finally decided to go for the kill with a series of new business techniques and an Initial Public Offering (IPO) worth \$2.7 billion (Rs 12,150 crore).

In a letter to the Securities and Exchange Commission, they said, "The standard structure of public ownership may jeopardise the independence and focused objectivity that have been most important in Google's past success, and that we consider most fundamental for its future. Therefore, we have designed a corporate structure that will protect Google's ability to innovate and retain its most distinctive characteristics." In effect, subscribers to the IPO will be offered Class A shares that carry a single vote per share, while insiders to the company will retain a right of ten votes per share.

Though a serious search engine, Google is notorious for its sense of humour, visit <http://www.google.com/intl/xx-piglatin/> to get an idea.

Sergey Brin and Larry Page
FOUNDERS, GOOGLE.COM

